

```
1 clc;
2 clear;
3 close all;
4
5 % Antenna Array Parameters
6 antennas = [2 4 8 16 32];
7
8 % Beamforming Gain (dB)
9 gain = 10*log10(antennas);
10
11 % Beam Width (degrees)
12 beamwidth = [90 45 22.5 11.25 5.625];
13
14 % Main Lobe Direction (degrees)
15 theta = [-60 -30 0 30 60];
16
17 % Constant Normalized Gain
18 normalized_gain = [1 1 1 1 1];
19
20 figure;
21
22 % -----
23 % Graph 1: Beamforming Gain
24 % -----
25 subplot(3,1,1)
26
27 bar(antennas, gain)
28 grid on
29 xlabel('Number of Antenna Elements')
30 ylabel('Beamforming Gain (dB)')
31 title('Beamforming Gain for Different Antenna Array Sizes')
32
33 % -----
34 % Graph 2: Beam Width
35 % -----
36 subplot(3,1,2)
37
38 plot(antennas, beamwidth, 'o-', 'LineWidth', 2)
39 grid on
40 xlabel('Number of Antenna Elements')
41 ylabel('Beam Width (Degrees)')
42 title('Beam Width for Different Antenna Array Sizes')
43
44 % -----
45 % Graph 3: Main Lobe Direction
46 % -----
47 subplot(3,1,3)
```

